VVU EARTH TECH

Assembly & Prototype Development Lifecycle Spec Guide

FOR OUTREACH DISTRIBUTION, SCOPING & SALES FRAMEWORK STRATEGIES

Target Audience: Prototype Development Teams, Assembly Engineers, QA Specialists

CLASSIFICATION: Outreach Distribution — Scoping & Sales Framework Strategy

Content Scope: Technical, Operational, and Strategic — Excludes Legal & Financial

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1. Prototype Lifecycle Overview

The VVU EARTH TECH Hydro-Gateway prototype follows a 6-phase lifecycle from concept through production readiness. Each phase has defined objectives, deliverables, and quality gates. The Epistemic DAG Runtime provides immutable evidence tracking at every phase — all test results, inspection data, and acceptance decisions are recorded as Facts in the Fact Log.

Phase	Name	Duration (est.)	Key Deliverable	Quality Gate
P0	Concept & Specification	4-8 weeks	24 parametric constraints + interfaces	Specification review + approval
P1	Design & Fabrication Planning	6-12 weeks	Design package (CAD, BOM, process plan)	Design review + fabrication readiness
P2	Prototype Build & FAI	8-16 weeks	Functional prototype + FAI results	All 24 constraints verified
P3	Integration & System Test	4-8 weeks	Integrated test report + telemetry verified	End-to-end Epistemic Runtime flow
P4	Field Trial (Cape Town)	12-24 weeks	Field trial data + performance analysis	Specification met in real conditions
P5	Production Readiness	4-8 weeks	Production process + QA procedures + supply chain	Production readiness review PASS

2. Lifecycle Phase Definitions

P0 — Concept & Specification

- **Objective:** Define all specifications, parametric constraints, interfaces, and performance requirements.
- **Activities:** Requirements analysis, constraint derivation, interface definition, risk identification.
- **Deliverables:** Specification document (24 parametric constraints), interface control document, risk register.
- **Quality Gate:** Specification review by engineering lead + independent review; approval recorded as Fact.

P1 — Design & Fabrication Planning

- **Objective:** Complete detailed design, material selection, and fabrication process planning.
- **Activities:** CAD design, material specification, BOM creation, fabrication process development, DFA/DFM analysis.
- **Deliverables:** Design package (CAD files, BOM, process plan, assembly sequence), design analysis report.
- **Quality Gate:** Design review; all dimensions verified against constraints; DFA/DFM analysis PASS.

P2 — Prototype Build & FAI

- **Objective:** Fabricate first article, assemble, calibrate, and verify against all 24 parametric constraints.
- **Activities:** Component fabrication, PCB assembly, sensor calibration, enclosure assembly, functional test.
- **Deliverables:** Functional prototype, FAI results (all 24 constraints), calibration records.
- **Quality Gate:** FAI PASS — all 24 parametric constraints met; results SHA-256 signed and recorded as Facts.

P3 — Integration & System Test

- **Objective:** Connect prototype to Epistemic Runtime and verify end-to-end telemetry flow.
- **Activities:** Runtime integration, telemetry flow verification, 5-Pass Evidence Compiler test, MMR indexing test.
- **Deliverables:** Integrated system test report, telemetry verification evidence, MMR root hashes.
- **Quality Gate:** End-to-end telemetry verified: acoustic event → Fact Log → MMR inclusion proof.

P4 — Field Trial (Cape Town Pilot)

- **Objective:** Deploy prototype at Cape Town pilot site and verify performance under real conditions.

• **Activities:** Site installation, commissioning, continuous monitoring, data collection, performance analysis.

• **Deliverables:** Field trial data (recorded as Facts), performance analysis report, environmental compliance evidence.

• **Quality Gate:** Performance meets specification in real Cape Town conditions; all Facts SHA-256 verified.

P5 — Production Readiness

• **Objective:** Finalize fabrication process, QA procedures, supply chain, and production documentation.

• **Activities:** Process optimization, QA procedure finalization, supply chain qualification, production documentation.

• **Deliverables:** Production process specification, QA manual, supply chain qualification report.

• **Quality Gate:** Production readiness review — all gates PASS; documentation SHA-256 signed.

3. Assembly Procedures & Sequences

The Hydro-Gateway assembly follows a defined sequence with quality gates at each step. All assembly data is recorded as Facts in the Epistemic Runtime.

Step	Operation	Tools/Equipment	Quality Gate	Evidence Recording
1	PCB preparation + ESD-safe handling	ESD workstation, magnification	Visual inspection (IPC-A-610)	Inspection Fact
2	Component placement + soldering	Solder station, reflow oven	Electrical test + X-ray (if BGA)	Test Fact
3	Conformal coating (specified areas)	Coating equipment, UV cure	Coating thickness verification	Verification Fact
4	Firmware loading + verification	Programming station	Firmware version + checksum verified	Checksum Fact
5	Sensor installation + calibration	Calibration fixture, reference standard	SNR ≥ 45 dB, freq ±3 dB	Calibration Fact
6	Communications module installation	RF test station	Network registration + data TX	Communications Fact
7	Enclosure assembly + seal	Torque tools, seal fixture	IP67 pressure test (1m, 30min)	Seal Verification Fact
8	Integration test (end-to-end)	Epistemic Runtime test bench	Acoustic → telemetry → Fact Log	Integration Fact
9	Final inspection + packaging	Inspection station	All 24 constraints verified	Final Inspection Fact

4. Component Integration Testing

Each component is tested individually before integration. Test results are recorded as Facts and indexed in the MMR for immutable evidence trails.

Component	Test	Method	Pass Criteria	Evidence
Acoustic Sensor	Sensitivity + frequency	Lab calibration vs reference	SNR ≥ 45 dB; ±3 dB flat	Calibration Fact (SHA-256 signed)
Processing Unit	Functional + throughput	DSP test suite	≥ 50 events/sec; no errors	Functional Test Fact
Communications Module	Network + transmission	Operator network test	Registration + data TX verified	Communications Test Fact
Power Management	Battery + solar	Load test + solar simulation	≥ 72h battery; ≥ 15Wh/day solar	Power Test Fact
Enclosure	IP67 + mechanical	Pressure test + dimensional	No ingress; dimensions within spec	Enclosure Test Fact
Secure Element	Ed25519 signing	Cryptographic test vector	Signature verification PASS	Crypto Test Fact

5. Epistemic Runtime Verification at Each Phase

The Epistemic DAG Runtime provides evidence tracking throughout the prototype lifecycle. Every quality gate, test result, and acceptance decision is recorded as an immutable Fact.

Lifecycle Phase	Epistemic Verification	Evidence Type	MMR Inclusion
P0 Concept	Specification review decision recorded	Policy (specification standard)	Root hash of spec review
P1 Design	Design review decision + constraint mapping	Fact (design parameters) + Proof	Root hash of design review
P2 Build	FAI results for all 24 constraints	Fact (each constraint measurement)	Root hash of FAI batch
P3 Integration	End-to-end telemetry flow verified	Fact (telemetry packet) + Proof (signature)	Root hash of integration test
P4 Field Trial	Performance data from Cape Town deployment	Fact (field measurements) + Projection	Root hash of field trial batch
P5 Production	Production readiness review decision	Policy (production standard) + Fact	Root hash of readiness review

6. Quality Gates & Acceptance Criteria

Gate	Phase	Criteria	Decision	Evidence
G0	P0	All specifications defined + constraints derived	PASS/FAIL	Specification review Fact
G1	P1	Design meets all constraints + DFA/DFM PASS	PASS/FAIL	Design review Fact
G2	P2	FAI: all 24 parametric constraints verified	PASS/FAIL	FAI measurement Facts (24)
G3	P3	End-to-end telemetry: acoustic → Fact Log	PASS/FAIL	Integration test Fact
G4	P4	Field performance meets specification	PASS/FAIL	Field trial Facts + analysis
G5	P5	Production process + QA + supply chain qualified	PASS/FAIL	Production readiness review Fact

■ ***FAIL at any gate does NOT terminate the project — it triggers a return to the previous phase for corrective action. The Epistemic Runtime records the failure as a Fact, enabling root cause analysis and evidence-based decision-making.***

7. Validation Milestone Tracking

Prototype lifecycle milestones are tracked alongside VVU-VAL-001 validation milestones in the Epistemic Runtime. Both hardware and software evidence share the same MMR.

Milestone	Name	Trigger	Actions
PM-0	Prototype Specification Approved	G0 PASS	Begin P1 design
PM-1	Design Package Complete	G1 PASS	Begin P2 build
PM-2	First Article Inspection PASS	G2 PASS (all 24 constraints)	Begin P3 integration
PM-3	Integration Test Verified	G3 PASS (end-to-end telemetry)	Begin P4 field trial
PM-4	Field Trial Complete	G4 PASS (Cape Town performance)	Begin P5 production readiness
PM-5	Production Readiness Approved	G5 PASS	Production launch

8. Cape Town Pilot Deployment Timeline

Cape Town serves as the alternative pilot municipality case study for the VVU EARTH TECH Hydro-Gateway deployment. Cape Town was selected following a governance review that identified operational risks at the originally proposed municipality. Cape Town's established water management infrastructure, progressive governance framework, and existing IoT deployment experience make it an ideal partner for validating municipal-grade infrastructure monitoring technology.

Phase	Activity	Duration	Dependencies
P0	Cape Town site survey + requirements	4-8 weeks	Municipal partnership agreement
P1	Cape Town-specific design adaptation	6-12 weeks	Site survey data + specifications
P2	Prototype fabrication + FAI	8-16 weeks	Design package + fabrication partner
P3	Cape Town integration + commissioning	4-8 weeks	Functional prototype + municipal IT
P4	Cape Town field trial + monitoring	12-24 weeks	Commissioned system + monitoring team
P5	Production readiness for Cape Town scale	4-8 weeks	Field trial results + supply chain

9. Handoff & Transition Procedures

- **Phase Transition:** Each phase transition is gated by a quality gate (G0-G5). Transition requires PASS decision recorded as Fact.
- **Documentation Handoff:** All design, fabrication, test, and field trial documentation is SHA-256 signed and transferred to the next phase team.
- **Evidence Handoff:** All Facts, Proofs, and MMR root hashes from previous phases are included in the next phase's evidence baseline.
- **Failure Return:** FAIL at any gate triggers return to the previous phase with failure analysis Facts and corrective action plan.
- **Knowledge Transfer:** Phase completion includes knowledge transfer session with the next phase team; attendance recorded as Fact.

10. Resource Acquisition Strategy

7-Track Resource Acquisition & Partnership Strategy: Track A — Municipal Partnerships: Active outreach to Cape Town and progressive municipalities for pilot deployment and validation partnerships. *Status: Active Outreach* **Track B — University Research Partnerships:** Engagement with South African and international universities for collaborative research, validation observation, and academic publication. *Status: Strategy* **Track C — Industry Integration:** Partnership development with water infrastructure, IoT, and municipal technology companies for integration, supply chain, and co-development. *Status: Strategy* **Track D — Public Funding:** Application to national and provincial technology innovation funds, water sector grants, and digital infrastructure programmes. *Status: Strategy* **Track E — Private Investment:** Structured engagement with impact investors, technology venture capital, and ESG-aligned funds. *Status: Strategy* **Track F — Sponsorship & Equipment:** Outreach to hardware manufacturers, cloud providers, and technology sponsors for equipment, infrastructure, and in-kind support. *Status: Strategy* **Track G — Community & Open Source:** Building developer community, open-source contributors, and civic technology networks for grassroots validation and adoption. *Status: Strategy*

Execution Principle: VVU EARTH TECH advances through multiple independent pathways in parallel. Every activity is subject to formal governance review. Progress is measured by verified engineering deliverables, not announcements. Where a pathway encounters obstacles, alternative pathways continue without delay. This parallel execution model ensures continuous forward progress regardless of individual pathway outcomes.

Communications Policy: VVU EARTH TECH follows a staged disclosure framework: (1) Engineering progress is communicated only after verification — not before. (2) Outreach is structured in three stages: Evidence Publication → Personalized Outreach → General Social & Press. (3) Sensitive operational details are shared on a need-to-know basis. (4) Mass-blast communication is structurally impossible — the outreach engine enforces sequential stage gates.